

Bankhead National Forest, AL, AMF3 (Main Site)
Atmospheric Radiation Measurement User Facility



ARM Data Quality Report (DQR) Table

Datastream	DQR	Quality	Subject	Start Date	End Date
bnfaerisummaryM1.b1	D250314.4	Suspect	Interferometer Skipping Fringes	2024-11-18T00:00:00	2025-02-12T17:00:00
bnfceilM1.b1	D251215.3	Missing	Missing Data	2025-11-08T12:41:10	2025-11-10T16:21:10
bnfceilM1.b1	D250509.8	Missing	Missing Data due to Power Outage	2025-03-04T23:59:52	2025-03-06T00:00:03
bnfceilM1.b1	D250512.10	Missing	No Data	2025-04-23T20:50:15	2025-04-25T00:00:06
bnfhslM1.a1	D260226.1	Missing	No Data	2026-01-25T00:00:00	2026-01-27T23:59:59
bnfhslM1.a1	D251124.2	Missing	Missing Data	2025-11-08T12:45:00	2025-11-10T15:00:00
bnfhslM1.a1	D250919.6	Missing	Missing Data	2025-05-14T20:00:07	2025-05-15T16:29:59
bnfhslM1.a1	D250811.1	Missing	No Data Available	2025-06-17T14:45:00	2025-07-27T00:45:00
bnfldM1.b1	D250122.8	Missing	Missing Data	2024-12-29T04:36:07	2025-01-07T15:14:59
bnfldM1.b1	D251215.8	Missing	Missing Data	2025-11-08T12:40:07	2025-11-12T14:30:59
bnfldM1.b1	D251202.5	Missing	No Data	2025-11-08T12:40:07	2025-11-12T14:30:59
bnfmawsM1.b1	D260317.18	Suspect	MAWS HMP-155 Reporting Questionable Data	2025-12-14T15:03:52	2026-03-04T23:59:59
bnfmawsM1.b1	D250319.1	Missing	Missing Data	2025-02-23T23:59:07	2025-02-25T00:00:04
bnfmfrsr7nchM1.b1	D250527.13	Incorrect	Incorrect Data	2024-12-23T12:52:00	2025-01-17T06:00:00
bnfmfrsr7nchM1.b1	D260209.3	Incorrect	Incorrect Data	2025-10-19T12:16:40	2026-01-21T21:52:00
bnfmfrsr7nchM1.b1	D250613.1	Incorrect	Incorrect Data	2025-03-27T13:51:20	2025-05-28T16:30:00
bnfmfrsr7nchM1.b1	D250929.5	Incorrect	Incorrect Data	2025-09-16T22:38:00	2025-09-19T23:56:20
bnfmfrsr7nchM1.b1	D250527.14	Missing	Missing Data	2025-01-29T20:52:47	2025-03-27T13:51:19
bnfmplpolfM1.b1	D241218.2	Missing	Missing Data	2024-12-09T00:00:00	2024-12-18T17:25:00
bnfmplpolfM1.b1	D241216.2	Missing	Missing Data	2024-09-27T13:46:24	2024-10-03T19:03:19
bnfmplpolfM1.b1	D250425.2	Missing	No Data	2025-04-20T13:20:17	2025-04-21T13:51:16
bnfmplpolfM1.b1	D260318.3	Missing	Data Missing	2026-03-16T05:42:56	2026-03-18T14:51:12
bnfmplpolfM1.b1	D250324.5	Incorrect	Data Incorrect	2025-02-24T19:00:05	2025-03-06T17:39:25
bnfmplpolfM1.b1	D250717.12	Suspect	Suspect Measurements	2025-07-03T21:00:00	2025-07-07T15:30:00
bnfmwr3cM1.b1	D250408.10	Missing	Missing Data	2025-04-07T23:07:35	2025-04-10T15:29:15
bnfmwr3cM1.b1	D250218.1	Missing	Missing Data	2025-03-05T01:00:00	2025-03-06T11:36:49
bnfmwr3cM1.b1	D251201.1	Missing	Missing Data	2025-11-14T13:00:00	2025-11-17T17:22:31
bnfmwr3cM1.b1	D260309.2	Missing	Missing Data	2026-01-23T17:00:00	2026-01-28T15:13:54
bnfmwr3cM1.b1	D250617.1	Missing	Missing Data	2025-05-19T07:00:00	2025-05-20T12:30:46
bnfmwr3cM1.b1	D250211.3	Missing	Missing Data	2024-10-01T00:00:00	2025-01-02T12:01:21

ARM Data Quality Report (DQR) Table

Datastream	DQR	Quality	Subject	Start Date	End Date
bnfmwr3cM1.b1	D250813.2	Missing	Missing Data	2025-08-08T09:00:00	2025-08-19T12:12:21
bnfmwr3cM1.b1	D260325.3	Missing	Missing Data	2026-02-28T20:00:00	2026-03-02T12:12:49
bnfmwr3cM1.b1	D260604.1	Missing	Missing Data	2026-05-05T19:59:57	2026-05-11T16:19:26
bnfmwr3cM1.b1	D250625.1	Missing	Missing Data	2025-06-20T16:00:00	2025-06-22T19:15:15
bnfmwr3cM1.b1	D250922.3	Missing	Missing Data	2025-09-04T11:00:00	2025-09-08T13:54:40
bnfmwr3cM1.b1	D250214.3	Incorrect	Incorrect Values	2025-01-13T00:00:00	2025-02-13T00:00:00
bnfmwr3cM1.b1	D250211.2	Suspect	Calibration Uncertain	2024-09-04T12:00:00	2024-09-18T23:59:59
bnfrlM1.a0	D251029.1	Incorrect	Incorrect Data due to Low Energy Level	2025-08-30T00:00:00	2025-10-06T15:28:36
bnfrlM1.a0	D260508.4	Incorrect	Incorrect Data	2026-05-06T21:21:32	2026-05-07T13:22:23
bnfrlM1.a0	D260303.13	Incorrect	Incorrect Data	2026-02-24T14:00:00	2026-02-24T17:19:02
bnfrlM1.a0	D250415.7	Missing	Missing Data	2025-04-12T14:44:28	2025-04-14T14:37:47
bnfrlM1.a0	D250407.2	Missing	Missing Data	2025-03-26T14:44:20	2025-03-28T14:59:19
bnfrlM1.a0	D260508.5	Missing	Missing Data	2026-05-07T13:22:23	3001-01-01 00:00:00
bnfrlM1.a0	D250807.8	Missing	Missing Data	2025-07-14T14:54:08	2025-07-15T21:58:41
bnfrlM1.a0	D260303.11	Missing	Missing Data	2026-02-24T17:19:02	2026-03-19T21:54:12
bnfrlM1.a0	D250224.4	Missing	Missing Data	2025-01-10T16:51:19	2025-01-13T18:25:07
bnfrlM1.a0	D251007.2	Missing	Missing Data	2025-09-04T11:31:20	2025-09-09T15:17:00
bnfrlM1.a0	D251109.2	Missing	Missing Data	2025-10-06T15:28:36	2025-10-29T21:53:19
bnfrlM1.a0	D250717.3	Missing	Missing Data	2025-06-26T09:31:29	2025-06-27T14:11:35
bnfrlM1.a0	D250827.2	Missing	Missing Data	2025-08-02T04:06:51	2025-08-04T14:39:18
bnfrlM1.a0	D250328.10	Missing	No Data	2025-02-16T05:48:09	2025-02-20T15:24:16
bnfrlM1.a0	D241112.2	Missing	Missing Data	2024-12-25T14:51:37	2024-12-31T16:42:10
bnfrlM1.a0	D260202.3	Missing	Missing Data	2026-01-30T23:22:12	2026-02-02T19:03:26
bnfrlM1.a0	D250327.2	Missing	Missing Data	2025-03-15T09:32:10	2025-03-17T14:23:58
bnfrlM1.a0	D250717.2	Missing	Missing Data	2025-06-07T18:35:42	2025-06-09T14:36:22
bnfrlM1.a0	D250519.2	Missing	Missing Data	2025-04-20T13:20:18	2025-04-21T14:54:51
bnfrlM1.a0	D260202.2	Missing	Missing Data	2026-01-22T13:44:33	2026-01-27T16:57:59
bnfskyrad60sM1.b1	D240929.3	Incorrect	Incorrect Data	2025-06-22T10:37:54	2025-06-22T12:16:00
bnfskyrad60sM1.b1	D250819.1	Incorrect	Incorrect Data	2025-08-14T19:51:00	2025-08-22T15:33:00
bnfskyrad60sM1.b1	D251212.1	Incorrect	Incorrect Data	2025-12-09T18:29:00	2025-12-10T14:05:00

ARM Data Quality Report (DQR) Table

Datastream	DQR	Quality	Subject	Start Date	End Date
bnfskyrad60sM1.b1	D241112.3	Incorrect	Data Incorrect due to Reversed Signal Polarity	2024-10-01T00:00:00	2024-10-24T18:27:00
bnfskyrad60sM1.b1	D251104.2	Suspect	Questionable Data	2025-10-10T13:18:00	2025-10-10T14:04:59
bnfskyrad60sM1.b1	D260408.3	Suspect	Questionable Data	2026-04-05T15:42:00	2026-04-05T23:29:59
bnfskyrad60sM1.b1	D241231.2	Suspect	Suspect Data due to Instrument/Environmental Conditions	2024-11-20T15:20:00	2024-11-20T16:20:00
bnfsondewnpnM1.b1	D250107.9	Missing	No SONDE Launches	2025-01-29T00:00:00	2025-01-29T23:59:59
bnfsondewnpnM1.b1	D250205.2	Missing	No Launches	2025-02-10T00:00:00	2025-02-11T23:59:59
bnfsondewnpnM1.b1	D260225.1	Missing	Missing Data	2026-01-25T00:00:00	2026-01-27T23:59:59
bnfvdisM1.b1	D250611.2	Missing	Missing Data	2025-06-14T00:00:00	2025-06-15T23:59:59
bnfvdisM1.b1	D241126.3	Missing	Missing Data	2024-10-01T00:00:00	2024-11-19T23:59:59
bnfvdisM1.b1	D260513.1	Missing	No Data	2026-05-01T00:00:00	2026-05-01T23:59:59
bnfvdisM1.b1	D260113.1	Missing	Missing Data	2025-12-11T00:00:00	2025-12-11T23:59:59
bnfvdisM1.b1	D250207.7	Missing	Missing Data	2025-01-28T00:00:00	2025-01-28T23:59:59
bnfvdisM1.b1	D250320.2	Missing	Missing Data	2025-03-28T00:00:00	2025-04-01T23:59:59
bnfvdisM1.b1	D260323.9	Missing	Missing Data	2026-02-19T00:00:00	2026-02-22T23:59:59
bnfvdisM1.b1	D260212.1	Missing	Missing Data	2026-01-26T00:00:00	2026-01-26T23:59:59
bnfvdisM1.b1	D250411.7	Missing	Missing Data	2025-04-07T00:00:00	2025-04-07T23:59:59
bnfvdisM1.b1	D251014.4	Missing	Missing Data	2025-10-10T00:00:00	2025-10-14T23:59:59
bnfvdisM1.b1	D260601.1	Missing	No Data	2026-05-22T00:00:00	2026-05-26T23:59:59
bnfvdisM1.b1	D251002.4	Missing	Missing Data	2025-09-18T00:00:00	2025-09-22T23:59:59
bnfvdisM1.b1	D251119.1	Missing	Missing Data	2025-11-08T00:00:00	2025-11-09T23:59:59
bnfvdisM1.b1	D260401.1	Missing	No Data	2026-03-17T00:00:00	2026-03-18T23:59:59
bnfvdisM1.b1	D260402.2	Missing	No Data	2026-03-25T00:00:00	2026-03-26T23:59:59
bnfvdisM1.b1	D241204.3	Missing	Missing Data	2024-11-22T00:00:00	2024-11-23T23:59:59
bnfvdisM1.b1	D251118.6	Missing	No Data	2025-10-29T00:00:00	2025-10-29T23:59:59
bnfvdisM1.b1	D251202.4	Incorrect	Incorrect Data	2025-11-26T00:00:00	2025-11-28T23:59:59
bnfwbpluvio2M1.a1	D251014.5	Suspect	Questionable Data	2025-06-22T00:00:00	2025-09-12T23:59:59
bnfwbpluvio2M1.a1	D250320.5	Suspect	Suspect Data	2025-03-05T01:00:00	2025-03-05T03:00:00
bnfwbpluvio2M1.a1	D250425.10	Suspect	Suspect Data	2025-04-09T15:00:00	2025-04-09T17:00:00
bnfwbpluvio2M1.a1	D250123.8	Suspect	Suspicious Data	2024-12-29T01:00:00	2024-12-29T05:00:00
bnfwbpluvio2M1.a1	D250728.2	Incorrect	Incorrect Data	2025-06-26T00:00:00	2025-07-14T23:59:59

ARM Data Object Identifier (DOI) Table

Instrument	DOI
AERI	Gero, J., Garcia, R., Hackel, D., Ermold, B., & Gaustad, K. Atmospheric Emitted Radiance Interferometer (AERICH1), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1989299
CEIL	Zhang, D., Morris, V., & Ermold, B. Ceilometer (CEIL10M), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1497398
DL	Newsom, R., Shi, Y., & Krishnamurthy, R. Doppler Lidar (DLCAL1), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1025183
GNDRAD	Sengupta, M., Habte, A., Andreas, A., Reda, I., Jaker, S., Xie, Y., Yang, J., Gotseff, P., Kutchenreiter, M., & Shi, Y. Ground Radiometers on Stand for Upwelling Radiation (GNDRAD60S), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1377837
HSRL	Bambha, R., Eloranta, E., Garcia, J., Razenkov, I., & Ermold, B. High Spectral Resolution Lidar (HSRL5S), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/2574248
IRT	Shi, Y., Howie, J., Goldberger, L., & Morris, V. Infrared Thermometer (IRT), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1329238
LDIS	Zhu, Z., Gonçalves Morais, F., & Shi, Y. Laser Disdrometer (LD), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1973058
MAWS	Keeler, E., Kyrouac, J., & Ermold, B. Automatic Weather Station (MAWS), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1162061
MET	Kyrouac, J., Shi, Y., & Tuftedal, M. Surface Meteorological Instrumentation (MET), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1786358
MFRSR	Hodges, G., Sheffer, B., Herrera, C., & Ermold, B. Multifilter Rotating Shadowband Radiometer (MFRSR7NCH), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1429369
MPL	Muradyan, P., Cromwell, E., Koontz, A., Coulter, R., Flynn, C., Ermold, B., & OBrien, J. Micropulse Lidar (MPLPOLFS), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1320657
MWR3C	Cadeddu, M., Gibler, G., Koontz, A., & Tuftedal, M. Microwave Radiometer, 3 Channel (MWR3C), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1025248
RL	Newsom, R., Bambha, R., & Chand, D. Raman Lidar (RL), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1996909

ARM Data Object Identifier (DOI) Table

Instrument	DOI
SKYRAD	Sengupta, M., Habte, A., Andreas, A., Reda, I., Jaker, S., Xie, Y., Yang, J., Gotseff, P., Kutchenreiter, M., & Shi, Y. Sky Radiometers on Stand for Downwelling Radiation (SKYRAD60S), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1377836
SONDE	Keeler, E., Burk, K., & Kyroutac, J. Balloon-Borne Sounding System (SONDEWNPN), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1595321
VDIS	Zhu, Z., & Gonçalves Morais, F. Video Disdrometer (VDIS), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1992988
WB	Zhu, Z., Wang, D., Jane, M., Cromwell, E., Sturm, M., Irving, K., & Delamere, J. Weighing Bucket Precipitation Gauge (WBPLUVIO2), 2025-1-01 to 2025-12-31, Bankhead National Forest, AL, USA; Long-term Mobile Facility (BNF), Bankhead National Forest, AL, AMF3 (Main Site) (M1). Atmospheric Radiation Measurement (ARM) User Facility. https://doi.org/10.5439/1338194